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Analysis of Community and Individual Resilience to Tidal Flood Disasters in Sayung District, Demak, Indonesia

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Abstract: Tidal flooding in Sayung District, Demak Regency, provides concrete evidence of the impacts of climate change and land subsidence in the northern coastal areas of Java, Indonesia. This study aims to analyze the level of community and individual resilience in facing tidal flood disasters using secondary data. Data were obtained from scientific publications, institutional reports, and disaster statistics from the National Disaster Management Agency (BNPB) and Statistics Indonesia of Demak Regency (BPS Demak Regency). The analysis focused on three main components hazard, vulnerability, and capacity which were then standardized using z-score transformation and combined to construct a composite resilience index. The results indicated that the level of community and individual resilience in Sayung District was generally moderate, with a total resilience index value of −7.944. 70% villages had moderate resilience. Village Loireng, Jetaksari, and Pilangsari were classified as highly resilient, while three villages were classified as having low resilience (Gemulak, Timbulsloko, and Surodadi). Social resilience is reflected in strong community cooperation, although prolonged exposure to tidal flooding shows signs of declining adaptive engagement among residents. Economic resilience is largely suppported by the continued viability of fisheries and aquaculture livelihoods, which remain central to local adaptive strategies. Meanwhile, environmental resilience is weakened by mangrove degradation and land subsidence. From an individual perspective, levels of disaster knowledge and social support contribute positively to adaptive capacity despite risk normalization behavior.

Keywords: Community resilience; Individual resilience; Tidal flooding; Climate change

1. Introduction

The phenomenon of global climate change has triggered an increase in the frequency and intensity of hydrometeorological disasters in various regions of the world, including Indonesia. One of the most obvious impacts of climate change in coastal areas is tidal flooding, that is the rise in sea levels that inundates land due to high tides or land subsidence (Ruan et al., 2024; Sarasadi & Rudiarto, 2021; Suratman et al., 2023). Sayung District in Demak Regency is one of the areas most affected by this phenomenon. In the last two decades, this area experienced an increase in the frequency of tidal flooding, which has caused damage to the infrastructure, loss of livelihoods, and a decline in the quality of community life (Asrofi et al., 2017). This condition requires systematic efforts to assess and strengthen the resilience of communities and individuals to these disasters.

Resilience in the context of disaster is not only understood as the ability of a community to survive a disaster, but also the ability to adapt and recover promptly after a disaster occurs (Bakarbessy, 2024; Lubis, 2019; Lv et al., 2024). In the local context, community resilience in coastal areas such as Sayung is shaped by various factors, including socio-economic conditions, education, social solidarity, institutional support, and knowledge of disasters. In addition to community aspects, individual resilience has a significant role because a person's decisions, risk perceptions, and adaptive actions can determine the level of safety and sustainability of life after a disaster (Adger et al., 2011; DeMello et al., 2020; Mattedi et al., 2024). Several previous studies emphasized the importance of community resilience in reducing risks of disasters in coastal areas. For example, research by Isrofi & Gunawan

(2025) showed that coastal communities in northern Central Java have developed community-based adaptation strategies, such as self-help house elevation, floating reading houses, simple embankment construction, and livelihood diversification.

However, most existing studies on tidal flooding resilience in coastal areas primarily emphasized collective or community-level adaptation strategies rather than examining resilience at the individual level. For instance, several coastal adaptation studies in northern Central Java focused on community-based initiatives such as collective infrastructure adaptation, livelihood diversification, and social cooperation mechanisms (Isrofi & Gunawan, 2025). While these studies successfully demonstrated the importance of communal capacity in disaster response, they provided limited analysis of how individual factors such as risk perception, psychological readiness, decision-making processes, and personal adaptive capacity shaped outcomes of resilience. As a result, the micro-level dynamics that influence whether households adopt adaptive actions or remain vulnerable are insufficiently understood. This limitation creates a significant analytical gap, as sustainable disaster preparedness depends not only on collective structures but also on the resilience capacities of individuals who ultimately make adaptation decisions within risk-prone environments (Luthar et al., 2000).

Previous studies highlighted the importance of resilience in reducing the impacts of disasters, particularly in coastal and flood-prone regions. Research has shown that community resilience plays a crucial role in strengthening collective adaptive capacity and disaster preparedness, while individual resilience influences behavioral responses and recovery processes at the household level. Studies conducted in coastal environments demonstrated that socio-economic vulnerability, environmental degradation, and institutional capacity significantly affected resilience outcomes in areas exposed to recurrent flooding events. Furthermore, integrated resilience assessments combining social, economic, and environmental dimensions are increasingly recognized as essential for understanding disaster risk dynamics and supporting sustainable adaptation strategies (Aldrich, 2012; Chapagain et al., 2025; Cutter et al., 2010; Nguyen et al., 2022).

Research on the Sendai Framework for Disaster Risk Reduction 2015–2030 confirmed that enhancing community resilience was a key pillar in achieving sustainable development (Lin et al., 2025). According to the United Nations Office for Disaster Risk Reduction (UNDRR, 2015), communities should have the capacity to prevent, respond to, and recover from disasters by strengthening social, economic, and institutional systems. The dimension of individual resilience plays an equally important role. This resilience encompasses a person’s psychological, emotional, and behavioral abilities in responding to disaster risks. Ungar (2011) explained that individual resilience was formed through the interaction of internal factors (such as self-efficacy, self-control, and disaster experience) and external factors (social support and a safe environment). In the Sayung context, understanding individual resilience is crucial because adaptation decisions, such as choosing to stay put, repairing a home, or seeking alternative livelihoods, often depend on risk perception and personal capacity. Therefore, this study compared and examined the relationship between individual resilience and community resilience so as to understand the socio-ecological resilience of coastal communities as a whole.

Sayung District has its own complexities as this area is located in a coastal lowland, of which most has experienced land subsidence of up to 6–10 cm per year (Dwiakram et al., 2020; Hardoyo et al., 2011). In addition, economic pressures and coastal urbanization have led some residents to remain in areas prone to tidal flooding, despite the increasing risk each year (Awah et al., 2024). This suggests that resilience is influenced not only by physical environmental factors, but also by the social and psychological decisions of individuals and communities. Therefore, a comprehensive assessment of resilience at the community and individual levels needs to be carried out to describe the actual state of preparedness of Sayung residents in facing tidal flooding.

In the context of national policy, the Indonesian government and the National Disaster Management Agency (BNPB) have mainstreamed the concept of resilience into its disaster risk reduction program, as outlined in the 2020–2044 Disaster Management Master Plan. This approach emphasizes the importance of strengthening community capacity as the first line of defense in facing disasters. However, implementation at the local level often faces various obstacles, including weak inter-agency coordination, limited micro-disaster data, and low levels of community awareness and participation, as highlighted by BNPB and the Pacific Disaster Center (BNPB, 2021; Pacific Disaster Center, 2020). Thus, a literature review focusing on community and individual resilience in Sayung is important to assess the extent to which the relevant policy resonates at the grassroots level.

Furthermore, tidal flooding in Sayung District is not only a physical or technical problem, but also a social and cultural one. The community’s economic reliance on the fisheries and coastal industry sectors creates a dilemma between economic needs and environmental safety. Several studies noted the phenomenon of “pseudo-resilience”, in which communities appear to be physically able to adapt to the conditions of tidal flooding, but actually experience a decline in quality of life, health degradation, and a loss of sense of security (Kusumastuti et al., 2022). This study sought to review the concept of resilience by examining the balance among the physical, social, economic, and psychological resilience of communities and individuals in coastal areas (Laidlaw & Percival, 2024).

From the initial literature review, there existed a research gap regarding the relationship between individual resilience and community resilience in coastal areas that are experiencing chronic disasters such as tidal flooding. Some studies only assessed structural and institutional aspects, while psychological and individual adaptive factors were often ignored (Rahmawati & Fariz, 2024). However, in the context of sustainable development and disaster risk reduction, an integrative approach was required to formulate human-centered resilience adaptation strategies. Therefore, it is imperative for this study to analyze and compare conceptually and empirically the extent to which community and individual resilience in Sayung District has been formed, as well as how the interaction between the two could strengthen its adaptive capacity to disasters like tidal flooding.

Theoretically, this article was based on the conceptual framework of community resilience and individual resilience developed by D'Ambrosio & Migheli (2025) and Ungar (2011), who emphasized the interrelationship among social, psychological, and structural factors in facing disasters. Based on secondary data, this article examined various relevant research findings and disaster reports, particularly those related to the socio-economic conditions, adaptation strategies, and risk perceptions of the Sayung community. The results in this study were expected to provide a comprehensive understanding of multi-level resilience (both individual and community levels) and inform local policies aimed at strengthening community-based disaster management systems.

Disaster resilience refers to the ability of an individual or community to survive, adapt, and recover from the effects of disasters. Khan et al. (2022) explained that increasing resilience was a crucial strategy in sustainable development because it strengthened the capacity of communities to face the risks of disasters. They developed the Comprehensive Disaster Resilience Index (CDRI), which encompasses nine key dimensions: economic stability, emergency workforce, agricultural development, human capital, digitalization, infrastructure, governance, social capital, and women's empowerment. This multidimensional approach is relevant for understanding the conditions of coastal communities in Sayung District, which encounter the threat of recurring tidal flooding. Their resilience is determined not only by environmental factors but also by local social, economic, and institutional capacities. Thus, this study aimed to assess and compare community and individual resilience levels across villages in Sayung District, with the use of a composite resilience index based on disaster and socio-economic data obtained from secondary sources.

2. Methodology

2.1 Research Design

This study employed a descriptive qualitative approach using secondary data analysis. This approach was chosen because the objective of this study was not to test empirical hypotheses, but rather to examine, identify, and synthesize various previous research findings related to community and individual resilience in the face of tidal flooding, particularly in Sayung District, Demak Regency (Lame, 2019). Conceptual references from previous disaster resilience studies were used to define analytical dimensions and indicators.

However, this study did not conduct a systematic literature review. Instead, the literature served as a theoretical foundation supporting indicator selection and interpretation. Village-level statistical and institutional data were analyzed to generate standardized resilience scores and enable comparative assessment among villages.

2.2 Sources and Types of Data

This study utilized secondary data obtained from governmental statistical publications and disaster management reports. The indicators were categorized into three analytical dimensions: hazard, vulnerability, and capacity. Hazard indicators included frequency of tidal floods, duration of inundation, and the extent of affected area obtained from disaster reports published by BNPB and the Demak Regency Disaster Management Agency (2021–2023).

Vulnerability indicators were derived from statistical publications issued by Statistics Indonesia of Demak Regency (BPS Demak Regency) for 2021–2023. Capacity indicators included availability of evacuation facilities, disaster preparedness programs, institutional support, and adaptive infrastructure compiled from village profiles and regional planning documents (2021–2023).

The secondary data in this study consisted of:

(1) Scientific publications (national and international journal articles, proceedings, and theses) relevant to the topics of community resilience, individual resilience, and tidal flood disasters on the north coast of Java.

(2) Policy documents and institutional reports were used as primary data sources, including the Disaster Management Master Plan of Demak Regency (2021–2023) issued by the Regional Disaster Management Agency of Demak Regency, as well as statistical reports published by Badan Pusat Statistik.

(3) Secondary statistical and spatial data, such as tidal flood incident data, socio-economic vulnerability levels, and the distribution of affected area, were obtained from BNPB reports and BPS Demak Regency publications for 2021–2023, as well as previous research publications.

The data were used to strengthen the literature analysis and provide an empirical picture of the condition of community and individual resilience in Sayung District. In the Indonesian Disaster Risk Index (IRBI) guidelines, BNPB used following equation (BNPB, 2021):

$$\text{Risk} = \frac{\text{Hazard } \times \text{Vulnerability}}{\text{Capacity}}$$

This confirms (1) risk increases when hazard (*H*) and vulnerability (*V*) are high and (2) risk decreases as capacity (C) increases. Therefore, resilience is the opposite. A society is more resilient with greater capacity and smaller vulnerability and hazard.

$$\text{Resilience} \propto \frac{C}{H \times V}$$

However, because sometimes risk (result of $H \times V/C$) is calculated separately, the formula can be simplified into another proportional function.

In field-level implementation or Geographic Information System (GIS) projects, several studies (including studies of community resilience in Indonesia, Bangladesh, Nepal, and the Philippines) simplify the formula to the form below:

$$\text{Resilience} = \frac{C}{V + R}$$

2.3 Data Collection

Data collection was conducted through documentation and data compilation from official institutional sources. Statistical data was obtained from BPS Demak Regency, while disaster-related information was collected from BNPB and regional disaster management reports. Village-level datasets corresponding to hazard, vulnerability, and capacity indicators were compiled and verified for consistency across the observation period from 2021–2023. Only indicators with complete and comparable data across villages were included in the analysis to ensure reliability of the resilience index calculation.

The stages of data collection were carried out through three systematic steps, namely:

(1) Literature Identification

A literature search was conducted through online databases, including Scopus and Web of Science, with the following keywords: “coastal community resilience”, “individual resilience”, “tidal flood”, “Demak”, and “Sayung”. The inclusion criteria included:

Articles published between 2010–2024 and contained discussions on community and individual resilience, or adaptation to tidal flooding, in the context of coastal areas of Indonesia, particularly Central Java.

(2) Literature Selection and Classification

All articles obtained were selected based on their relevance to the research topic and objectives. Articles that passed the selection stage were classified into three main categories: (1) literature on community resilience; (2) literature on individual resilience; and (3) literature on tidal flooding in Sayung or similar coastal areas.

(3) Data and Information Synthesis

Information from various sources of literature was then synthesized using a thematic analysis approach. The main themes identified included: (a) dimensions of community and individual resilience; (b) determinants of resilience; (c) adaptation strategies to tidal flooding; and (d) research gaps and opportunities for policy intervention.

2.4 Data Analysis

The analysis involved the construction of a composite resilience index using standardized quantitative indicators. Indicators were transformed into the same analytical direction before aggregation. Hazard and vulnerability indicators were inversely interpreted because higher values indicated greater risk, while capacity indicators retained positive interpretation. This ensured that higher index values consistently reflected higher resilience. All indicators were assigned equal weights to avoid subjective bias and to maintain analytical simplicity in the construction of composite index.

This research was conducted in Sayung Regency using secondary data obtained from the Demak Regency Central Statistics Agency. The hazard component was measured using numerical data consisting of the number of disaster events and the number of disaster victims in each village.

The vulnerability component was measured using several indicators, including social vulnerability, represented by population density; geographic vulnerability, represented by geographic location, area, and administrative boundaries; economic vulnerability, represented by agricultural land use for food crops, horticulture, plantations, and livestock (ha); and physical vulnerability, represented by the number of educational facilities, sports facilities, and health facilities exposed to disaster risk.

Meanwhile, the capacity component was measured using indicators related to disaster preparedness and mitigation infrastructure, including the availability of early warning systems, tsunami warning systems, safety equipment, evacuation signs, the construction and maintenance of disaster management infrastructure (rivers, canals, embankments, drainage systems, reservoirs, coastal protection structures, etc.), and village water reservoirs.

All indicator values obtained from the Demak Regency Central Statistics Agency statistical data were standardized using the resilience index (z-score) transformation method to ensure comparability between villages. The z-score was calculated by integrating standardized scores of hazards, vulnerability, and capacity dimensions following a proportional relationship of resilience calculation. For each component, the mean and standard deviation were calculated across all observational units. The standard deviation of component j was calculated as:

$$S_j = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{(n - 1)}}$$

Where, S_j is the standard deviation of component j, x_{ij} is the original value of component j for village i, $\bar{x}_j$ is the mean value of indicator j and n is the number of observations.

The z-score is calculated using following formula:

$$z_i = \frac{(x_{ij} - \bar{x}_j)}{S_j}$$

Village scores were obtained by aggregating standardized indicators within each dimension and computing the final composite index.

Resilience levels were categorized according to standardized range of scores: when $z > 0.5$, resilience is high; when $-0.5 \leq z \leq 0.5$, resilience is moderate; when $z < -0.5$, resilience is low. This classification facilitates comparison of resilience performance among villages.

Data analysis was carried out by combining two approaches:

(1) Thematic analysis

This approach was used to identify recurring patterns and themes from various literature. Thematic analysis was conducted following the stages of Braun & Clarke (2006), namely: (1) familiarization with the data; (2) initial coding; (3) theme search; (4) theme review; (5) defining and naming themes; and (6) preparing a synthesis report.

(2) Descriptive analysis of secondary data

Secondary data were analyzed descriptively to strengthen the literature findings. This analysis included trends of tidal flooding (frequency, duration, and extent), socio-economic conditions in the community, and the levels of exposure and vulnerability of Sayung District.

To enhance the reliability of the analysis, the data and literature used were verified by source triangulation, such as comparing findings from academic publications with institutional data and policy reports.

3. Results

3.1 Analysis Results of Community and Individual Resilience

The z-score calculation for the three main components, i.e., hazard, vulnerability, and capacity, determined the levels of community and individual resilience in 20 villages in Sayung District. These results are visualized in Figure 1, which shows variations in resilience among different regions.

In general, the z-score distribution presented in Figure 1 and Table 1 indicates that resilience levels across villages in Sayung District are predominantly concentrated within the moderate category. This interpretation is supported by the numerical distribution, where 14 out of 20 villages (70%) fell within the standardized range of $-0.5 \leq z \leq 0.5$. Three villages Loireng ($z = 1.188$), Jetaksari ($z = 0.709$), and Pilangsari ($z = 0.557$) exceeded the upper threshold and were classified as highly resilient, while three villages Gemulak ($z = -4.074$), Timbulsloko ($z = -2.118$), and Surodadi ($z = -1.048$) fell below the lower resilience threshold and were categorized as having low resilience.

Villages categorized as moderately resilient demonstrated relatively balanced relationships among vulnerability, capacity, and risk indicators rather than exceptionally strong adaptive capacity. For example, Sayung Village ($z = 0.233$) and Tambakroto ($z = 0.122$) showed resilience values close to the district average, indicating neither extreme vulnerability nor strong institutional capacity. This suggests that resilience in most villages remains adaptive rather than transformative.

The high resilience score observed in Loireng Village is directly reflected in its positive standardized value, indicating comparatively stronger capacity indicators relative to vulnerability and risk components. Similarly, Jetaksari and Pilangsari are also classified as high resilience villages, further reinforcing the presence of strong adaptive capacity in selected areas. Conversely, the substantially negative scores recorded in Gemulak and

Timbulsloko indicate that vulnerability and risk indicators outweigh existing adaptive capacities, as shown by their deviation from the district mean. This pattern is also evident in Surodadi, which is categorized as a low resilience village. Overall, the cumulative resilience score of −7.944 suggests that resilience conditions across Sayung District remain slightly below the standardized average, hence reinforcing the dominance of moderate resilience conditions rather than widespread high resilience.

From an individual perspective, the literature synthesis indicates that residents with better disaster knowledge, high adaptability, and strong social support tend to have greater psychological resilience. In the Sayung context, these factors are reflected in community groups that have adapted by building stilt houses, diversifying their work, or moving to safer locations while maintaining social ties within their original communities.

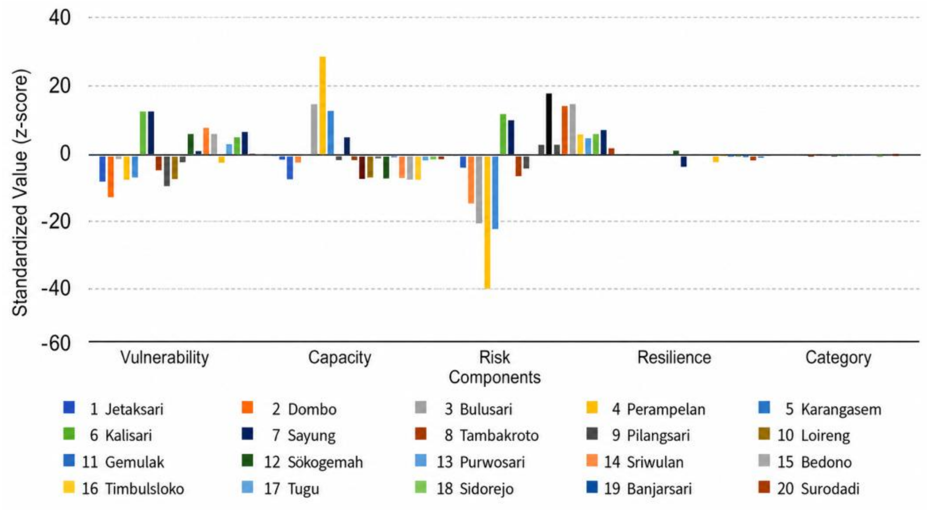

Figure 1. Standardized values of hazard, vulnerability, capacity, and resilience in Sayung District
Source: BPS Demak 2021–2023

Table 1. The *z*-scores of village resilience in Sayung District

No.	Village	Resilience Scores	Category
1	Jetaksari	0.709	High
2	Dombo	0.018	Medium
3	Bulusari	−0.748	Medium
4	Perampelan	−0.629	Medium
5	Karangasem	−0.468	Medium
6	Kalisari	−0.052	Medium
7	Sayung	0.233	Medium
8	Tambakroto	0.122	Medium
9	Pilangsari	0.557	High
10	Loireng	1.188	High
11	Gemulak	−4.074	Low
12	Sidogemah	−0.296	Medium
13	Purwosari	−0.276	Medium
14	Sriwulan	−0.324	Medium
15	Bedono	−0.352	Medium
16	Timbulsloko	−2.119	Low
17	Tugu	−0.171	Medium
18	Sidorejo	−0.122	Medium
19	Banjarsari	−0.095	Medium
20	Surodadi	−1.048	Low
Total resilience scores		−7.944	

Source: BPS Demak 2021–2023

The analysis of community resilience in Sayung District was conducted by considering three main variables: vulnerability, capacity, and risk. The resilience score was obtained through a *z*-score transformation of each variable, which was then combined into a single composite index. Overall, the total resilience score for Sayung District was recorded at −7.944, indicating that the overall resilience in the region is still classified as moderate to low.

3.2 Distribution of Resilience Between Villages

The calculations revealed significant variation in the level of community resilience across the 20 villages within the study area. Based on the categories, there were three main levels: high, medium, and low, with the following proportions: 3 village (15%) in high category, 14 villages (70%) in medium category, and 3 villages (15%) in low category. These categories were determined based on the interval of the resilience value of the *z*-score results, where positive values indicate resilience that is relatively better than average. In contrast, negative values indicate resilience that is below average.

(1) Villages with high resilience

Loireng was the village with the highest resilience score (1.188), indicating relatively strong community and institutional capacity, as well as lower levels of vulnerability and risk compared with other villages. In addition, Jetaksari (0.709) and Pilangsari (0.557) were also categorized as having high resilience, although their scores remain below that of Loireng. These findings suggest that while several villages demonstrate strong adaptive capacity, Loireng stands out as the most resilient. Its geographical location, which is more protected from tidal flooding, and good socio-economic support from the community, contribute to the resilience of this region (Ashrida et al., 2025).

(2) Villages with moderate resilience

Most of the Sayung area fell into the moderate resilience category, with resilience values ranging from 0.12 to 0.70. Some villages with moderate resilience included: Dombo (0.018), Bulusari (−0.748), Perampelan (−0.629), Karangasem (−0.468), Kalisari (−0.052), Sayung (0.233), Tambakroto (0.122), Sidogemah (−0.296), Purwosari (−0.276), Sriwulan (−0.324), Bedono (−0.352), Tugu (−0.171), Sidorejo (−0.122), Banjarsari (−0.095). These villages demonstrated a balance between capacity and vulnerability, but were not yet resilient enough to significantly mitigate the impact of risks (Zhai & Lee, 2024). For example, both Sayung and Tambakroto villages had relatively high vulnerability scores, but indicated moderate social and institutional capacity, thus being classified into the moderate category. This illustrates that the community in the central region of Sayung District possesses good adaptation capabilities, particularly in managing the disasters of tidal flooding that regularly occur.

(3) Villages with low resilience

Villages showing low resilience levels were Gemulak (−4.074), Timbulsloko (−2.118), and Surodadi (−1.048). Among these, Gemulak had the lowest resilience score, indicating the most critical condition. This was due to a combination of high vulnerability, significant risk, and low community capacity. The village's coastal location leads to high exposure to tidal flooding and seawater intrusion. In addition, limited adaptive infrastructure, such as embankments and drainage channels, as well as low community participation in mitigation activities, further weaken the overall resilience level.

3.3 General Interpretations and Policy Implications

Judging from the distribution of values, there were no villages with extreme resilience (very high or very low), but the negative trend in total resilience (−7.944) indicates that the collective resilience of Sayung District remains below the ideal average. The predominance of moderate resilience indicates that adaptive potential has not been optimally utilized. Local governments can use these results as a reference for prioritizing capacity building, particularly in community empowerment in coastal areas, such as Timbulsloko and Bedono. Besides, local governments can strengthen tidal flood control infrastructure and drainage systems and improve coordination between villages with early warning systems of disasters.

These results supported the importance of strengthening community-based disaster education, enhancing local institutional capacity, and implementing risk-based spatial planning. Villages with moderate resilience should be directed to become learning models for villages with low resilience, through a participatory and collaborative cross-sectoral approach. Research findings in Sayung District indicate low social and infrastructural capacity, which highlights the need to strengthen efforts in line with the first and second priorities of the Sendai Framework for Disaster Risk Reduction 2015–2030, particularly in improving disaster risk understanding and enhancing risk reduction governance at the local level. Strengthening village institutions, increasing disaster literacy, and integrating risk data are important steps in increasing the resilience of coastal communities (UNDRR, 2015).

4. Discussion

4.1 Social Resilience

The discussion presented in this section was grounded in the empirical findings obtained from the resilience index analysis in Sayung District. The dominance of moderate resilience levels across 70% of villages indicates that adaptive capacity exists but remains insufficient to offset vulnerability and risk pressures. Therefore, the following discussion interprets social, economic, environmental, and individual resilience dimensions specifically in relation to the observed village-level resilience scores. Although residents are accustomed to dealing with tidal flooding regularly, forms of cooperation and collective action remain important social capital in the face of disasters. However, the phenomenon of fatigue resilience or declining participation in adaptation activities is emerging in communities that have been exposed to tidal flooding for too long, as indicated by a decrease in participation in disaster-related activities (Chapagain et al., 2025; Rahmawati & Fariz, 2024). This indicates the need to revitalize social institutions to maintain community cohesion.

Social resilience in Sayung District is strongly sharped by the level of solidarity, participation, and social networks among residents. Research results indicated that the community continued to rely on cooperation to mitigate tidal flooding, such as building temporary embankments, raising house floors, and sharing resources during disasters. This aligns with the findings of Kusumastuti et al. (2022), who noted that social solidarity and trust among community members were crucial assets in building community resilience in the coastal areas of Semarang and Demak.

The decline in participation in adaptation activities is beginning to appear in frequently affected areas such as Gemulak and Timbulsloko. Communities experiencing tidal flooding almost every month are showing a decrease in motivation to participate in mitigation activities. This is supported by Rahmawati & Fariz (2024), who found that declining participation in adaptation activities reduced the effectiveness of community institutions in reducing risk. This condition reflects a decline in social adaptive capacity, in which the collective ability to respond to disasters begins to weaken due to repeated stress (UNDRR, 2023).

Conceptually, social resilience requires three main forms of capacity: Reactive, adaptive, and transformative. In Sayung, reactive capacity (the initial response to tidal flooding) is well established, as evidenced in cooperation and communication among residents. However, transformative capacity is the ability to create new and more resilient social systems as collective relocation or resilient village management is not yet optimal. Therefore, strengthening social resilience needs to be directed at revitalizing local institutions such as disaster preparedness groups, the Family Welfare Movement (PKK), and youth organizations (karang taruna) to enable them to initiate new adaptation innovations based on community participation.

4.2 Economic Resilience

The tidal floods in Sayung District damaged residents' homes, public facilities, rice fields, and fish ponds, which constitute their primary sources of livelihood. In response, the community utilized communal spaces as forums for deliberation to address the impacts of tidal flooding. These public spaces emerged through informal and semi-formal interactions, such as roadside gatherings (cangrukan), prayer rooms, stalls, and religious forums. During these deliberations, all residents actively expressed their opinions regardless of gender, resulting in inclusive and structured agreements. Importantly, these collective discussions played a crucial role in supporting economic resilience, as they facilitated joint decision-making on livelihood recovery strategies, resource allocation, and adaptive responses to sustain fisheries and agricultural activities amid recurring tidal flood risks. This is supported by previous findings reported by Masrohatun et al. (2025), who documented that agreed-upon activities, such as fundraising and tobacco planting, were successfully implemented.

Economic factors are the most important dimension of resilience in the Sayung community. Villages with multiple livelihoods (e.g., small businesses, trade, and non-fishing jobs) show greater resilience than villages that rely solely on the fisheries and aquaculture sectors. Economic reliance on coastal resources renders it difficult for some residents to move from areas prone to tidal flooding, thus increasing long-term risks (Hamid et al., 2021; Setyowati & Rachman, 2016). Economic resilience in Sayung District remains moderate, as it is marked by a high reliance on the fisheries and aquaculture sectors. Villages such as Bedono and Timbulsloko have experienced declining productivity due to seawater intrusion and the loss of aquacultural land, resulting in reduced household incomes. Economic reliance on coastal resources increases long-term economic risks for the Demak community due to the lack of livelihood diversification.

In contrast, villages with better economic diversification, such as Jetaksari, Pilangsari and Loireng, demonstrated relatively high resilience. This finding aligns with Khan et al. (2022), who emphasized that economic stability and income diversification were crucial factors in the CDRI. Communities with micro-enterprises, trade, or jobs in the non-coastal sector have greater financial capacity to recover from losses caused by tidal flooding.

Local economic conditions are shaped by access to institutional support. Government programs such as coastal micro, small, and medium enterprises (MSME) assistance, productive economic training, and labor-intensive projects from the Regional Disaster Management Agency (BPBD) remain sporadic. In fact, BNPB recommended strengthening a community-based adaptive economy (BNPB, 2021), where assistance was directed at increasing the entrepreneurial and microfinance capacity of villages. Thus, economic strengthening in Sayung requires not only financial assistance but also long-term policies that encourage the transformation of coastal economic structures into a more resilient model in response to climate change.

Research by Khan et al. (2022) showed that low-income countries tended to have low levels of resilience due to weak infrastructure, economic constraints, and inadequate institutional capacity. This situation is similar to that in Sayung District, where limited basic infrastructure, low community economic capacity, and suboptimal environmental management contribute to low resilience to tidal flooding. In addition, Khan et al. (2022) emphasized the importance of strengthening social capital and empowering women as essential elements in increasing the adaptive capacity of communities. In the Sayung context, social solidarity, cooperation, and the active involvement of women in preparedness activities are local potentials that can be developed to strengthen the resilience of coastal communities.

4.3 Environmental Sustainability

Environmental resilience remains weak due to damage to the coastal ecosystem and land subsidence reaching 6–10 cm per year (Hardoyo et al., 2011). Physical adaptation efforts, such as embankment construction, land reclamation, and house elevation, have been undertaken; however, most are reactive and short term in nature (Nugroho et al., 2025). In addition, mangrove degradation in the northern Sayung area reduces natural protection against tidal flooding, thus hampering the development of sustainable ecological resilience.

The environmental dimension is the most critical aspect of resilience in the Sayung community. Land subsidence of up to 10 cm per year and massive coastal abrasion caused the loss of more than 500 hectares of productive land in the last two decades (Hardoyo et al., 2011). Damage to the mangrove ecosystem in Bedono and Timbulsloko villages exacerbates exposure to tidal flooding, hence reducing the natural capacity of the areas to dampen tidal waves.

Environmental adaptation efforts have been undertaken, such as seawall construction, land reclamation, and mangrove planting; however, some of these efforts are reactive and unsustainable. Kusumastuti et al. (2022) noted that physical projects in Demak often failed to consider ecosystem dynamics, thereby creating new problems such as sedimentation and changes in current patterns. Meanwhile, the Intergovernmental Panel on Climate Change (IPCC) emphasized that environmental resilience was founded upon ecosystem-based adaptation (EbA); that is, ecosystem restoration and conservation as an integral part of disaster mitigation strategies (IPCC, 2023).

In the Sayung context, mangrove restoration, green belt formation, and integrated coastal space management are key steps in increasing ecological resilience. The successful implementation of Community-Based Mangrove Rehabilitation, as demonstrated in Bedono Village, can serve as a model for other villages. Thus, environmental resilience not only strengthens physical protection but also expands economic and social benefits for coastal communities (Abiyoga et al., 2018).

4.4 Individual Resilience

Individual resilience is closely linked to risk perception and personal adaptability. According to the literature, individuals with higher education, prior disaster experience, and strong social support tended to demonstrate higher levels of preparedness. However, some Sayung residents still exhibit risk normalization behavior, viewing tidal flooding as a common occurrence that does not require avoidance. This attitude indicates a gap between disaster knowledge and adaptive actions.

Individual resilience in Sayung District is closely related to residents’ risk perception, disaster knowledge, and social support. Based on secondary field analysis, individuals with higher education levels and prior disaster experience demonstrated greater preparedness, such as planning evacuation routes and storing important documents in secure locations. This aligns with Ungar (2011), who emphasized that individual resilience was determined not only by psychological factors but also by social interactions and a supportive environment.

However, most communities exhibit a normalization of risk, viewing tidal flooding as a routine part of life that does not require avoidance. This attitude makes it difficult to implement long-term adaptive measures, such as relocation or household risk planning. Rahmawati & Fariz (2024) and the National Disaster Management Agency (BNPB, 2021) noted that disaster literacy among communities on the north coast of Java remained low; therefore, efforts to increase individual resilience should focus on ongoing education about disasters.

The Disaster Resilient Village Program (Destana/Desa Tangguh Bencana) could be a vehicle for building individual resilience through preparedness training, evacuation simulations, and psychosocial support (Salman et al., 2024; Setiawan, 2025). The main factors that strengthen individual resilience are self-efficacy, social support, and the meaning of life after a disaster (Ferdinanto & Triyono, 2025). By increasing these capacities, the Sayung community is not only able to survive physically, but can also develop mental and emotional resilience against the risk of recurring tidal floods (Wahyudin, 2024).

4.5 Synthesis and Implication

The results of cross-dimensional analysis indicated that community and individual resilience in Sayung District was adaptive, implying the ability to adapt to tidal flooding without significant structural changes. The community could survive through cooperation, house elevation, and small-scale economic diversification, but has not yet achieved the transformative resilience conceptualized by Isrofi & Gunawan (2025). In general, villages with stronger social and economic capacities, such as Loireng, Pilangsari, and Jetaksari, showed higher resilience than coastal villages like Timbulsloko and Gemulak in experiencing environmental degradation. This pattern emphasizes the relationship among social, economic, and environmental dimensions in forming comprehensive resilience (Khan et al., 2022; Wen et al., 2023).

The implication is that increasing the resilience of the Sayung community requires an integrative strategy, which includes: (1) strengthen local social institutions such as disaster preparedness groups and village PRB (Pengurangan Risiko Bencana/Disaster Risk Reduction) forums; (2) diversify coastal economies to reduce reliance on vulnerable sectors such as fisheries; (3) advocate community-based environmental restoration through mangrove rehabilitation and adaptive spatial planning; and (4) improve individual literacy and psychosocial resilience through disaster education in schools and communities. This effort aligns with the Sendai Framework for Disaster Risk Reduction 2015–2030, which emphasizes the importance of risk governance and community participation as the foundation of sustainable resilience. If this cross-dimensional strategy is consistently implemented, Sayung can transform from an adaptive area into a resilient coastal community capable of facing the dynamics of climate change and future tidal flooding.

5. Conclusions

This study analyzed community and individual resilience in facing disasters of tidal floods in Sayung District, Demak Regency, using synthesis of secondary data and a resilience index approach based on hazard, vulnerability, and capacity dimensions. The findings indicated that overall resilience remained at a moderate level of index value −7.944, suggesting that communities possessed adaptive capabilities but had not yet reached transformative resilience to reduce long-term structural risks.

From a dimensional perspective, social resilience is strengthened by collective cooperation and social solidarity, although prolonged exposure to tidal flooding has resulted in declining participation in adaptation activities. Economic resilience remains vulnerable due to reliance on fisheries and aquaculture livelihoods, while environmental resilience represents the most critical challenge because of land subsidence, coastal abrasion, and mangrove degradation. At the individual level, disaster knowledge, psychological readiness, and social support contribute positively to adaptive capacity; however, risk normalization behavior continues to limit long-term preparedness.

Theoretically, this study contributes to disaster resilience scholarship by integrating community-level and individual-level resilience perspectives, thus demonstrating that resilience outcomes emerge from the interaction among social structures, economic resources, environmental conditions, and psychological capacities. This integrative perspective helps address the research gap in coastal disaster studies that often emphasize collective adaptation while overlooking individual adaptive dynamics.

Practically, the findings imply the need for integrated resilience strategies, including risk-informed spatial planning, coastal ecosystem restoration, livelihood diversification, and strengthened disaster education programs. Cross-sectoral collaboration among government institutions, communities, and educational organizations is essential to turn adaptive resilience into transformative resilience, to be aligned with the Sendai Framework for Disaster Risk Reduction 2015–2030.

This study is limited by its reliance on secondary data and literature-based synthesis, which may not fully capture dynamic behavioral responses at the household level. Future research is therefore recommended to employ primary data collection, a longitudinal approach, and a mixed-method design to examine how individual decision-making processes influence long-term transformation of resilience in disaster-prone coastal areas.

Author Contributions

Conceptualization, N.H. and I.V.A.; methodology, N.H.; software, N.H.; validation, N.H. and I.V.A.; formal analysis, I.V.A.; investigation, N.H.; resources, N.H.; data curation, N.H.; writing—original draft preparation, N.H.; writing—review and editing, I.V.A.; visualization, N.H.; supervision, I.V.A.; project administration, N.H.; funding acquisition, N.H. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that no generative artificial intelligence (AI) or AI-assisted technologies were used in the preparation of this manuscript. All writing, analysis, interpretation, and editing were conducted solely by the authors. The authors remain fully responsible for the accuracy, originality, and integrity of the manuscript content in accordance with ethical and publishing standards.

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